How FixServices Works and Finds Local Services

FixServices connects customers with home-service professionals who serve their town, taluka, or district. Customers select a location and category; the website finds matching service listings from approved providers and lets the customer submit a booking request.

> The local search is based on location details saved in provider profiles. It does not require a pincode. It is not GPS distance calculation unless a separate coordinate-based feature is added.

1. Complete working flow

Customer opens the website

The home page explains the service and directs visitors to browse services or create an account. Browsing can be public; booking requires a customer account.

Customer chooses a location

On the services page, the customer selects a district and optionally enters a taluka and town/village. The customer can also choose a category such as Plumbing, Electrical, Cleaning, Carpentry, Painting, or Appliance Repair. This version has no keyword/service-name search field.

System searches the database

PHP reads the selected filters, validates them, and runs a prepared SQL query. The query only returns active services from active categories and approved providers.

Matching listings appear

Each service card displays its title, category, description, provider name, town, taluka, district, price, optional rating, and a booking link. If nothing matches, the page displays a helpful no-results message.

Customer submits a booking

The customer opens a listing, chooses a date, enters the service address, adds optional notes, and submits. The server rechecks that the service is active and its provider is approved.

Provider responds

The request appears in the provider dashboard. The provider accepts or cancels it according to the project's rules. After finishing the work, the provider marks it completed.

Customer tracks the request

The customer's booking page displays their own bookings and their current statuses. A pending booking can be cancelled if the rules permit.

2. How local location matching works

When registering, a provider saves their location. For example:

Field	Example
State	Maharashtra
District	Kolhapur
Taluka	Karveer
Town / village	Kolhapur

These values are saved in the users table. The services table records what that provider offers. The services page joins the tables, allowing the system to filter by both service and provider location.

How the filters combine

- If district is selected, results must match that district.
- If taluka is also entered, results must match both district and taluka.
- If town is also entered, results must match district, taluka, and town.
- If a category is selected, results must belong to that category too.
- If a field is left blank, it does not restrict results for that field.

Example: selecting Kolhapur district, Karveer taluka, Kolhapur town, and Electrical category returns active electrical services from approved providers whose saved locations match those filters. Clearing the town field broadens the results while the remaining filters still apply.

3. Example: finding an electrician in Kolhapur

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Selected filters:
State: Maharashtra
District: Kolhapur
Taluka: Karveer
Town: Kolhapur
Category: Electrical

Search process:
1. Select active service listings.
2. Join each service to its category.
3. Join each service to its provider.
4. Exclude inactive categories.
5. Exclude unapproved or blocked providers.
6. Match district, taluka, and town.
7. Match the Electrical category.
8. Display matching service cards.
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If one provider is in Kolhapur town, another is elsewhere in Kolhapur district, and a third is in Pune, the full set of filters should return only records matching the selected district, taluka, and town. If the town filter is removed, results may include other matching towns within the selected taluka.

Important: a provider's registered town does not prove that they travel to surrounding towns. If providers serve multiple locations, add a separate service-area table or let providers specify multiple towns they cover.

4. Location matching is not GPS distance

This implementation compares location names such as district, taluka, and town. It does not know the customer's current GPS coordinates, calculate kilometres, or determine which provider is physically closest. A result can be described as a local-area match, not a measured nearest provider.

To implement actual nearest-provider search later, collect verified latitude/longitude coordinates for service areas and calculate distances or use an appropriate mapping service.

5. Provider approval and service listing

- A provider registers and enters contact and location details.
- The account is marked pending until an administrator reviews it.
- The administrator approves or blocks the provider.
- Approved providers can create listings with category, title, description, price, and active status.
- Only active listings in active categories from approved providers appear in customer search.
- Providers can edit only their own services; deactivated listings should disappear from search.

The registration page, login page, admin panel, and services query must use the same provider-approval column and status values. For example, choose provider_status with values such as pending, approved, and blocked and use it

consistently.

6. Booking lifecycle

Status	Meaning	Expected action
pending	Request submitted	Provider accepts or cancels
accepted	Provider agreed	Provider completes, or cancels if allowed
completed	Work finished	Normally final
cancelled	Booking cancelled	Normally final

Every status change must be checked on the server. Providers can only update bookings related to their own services. Customers can only view their own bookings and cancel only when the current status allows it.

7. What each page contributes

- **index.php:** introduces the service and links to service search, login, and registration.
- **register.php:** creates customer/provider accounts and saves location details.
- **login.php:** authenticates users and sends them to the correct role dashboard.
- **services.php:** filters by category, district, taluka, and town; no keyword field.
- **book.php:** validates a listing and creates a pending booking.
- **my_bookings.php:** displays the logged-in customer's own booking history.
- **provider_dashboard.php:** manages the provider's listings and their associated bookings.
- **admin.php:** approves or blocks providers, manages categories, and monitors bookings.
- **config.php:** provides database access, sessions, role checks, CSRF, escaping, and shared helpers.
- **includes/header.php / includes/footer.php:** keep navigation and page layout consistent.

8. Data quality makes local search useful

- Use a controlled district list to reduce spelling differences.
- Prefer dependent dropdowns: selecting a district loads its talukas, and selecting a taluka loads its towns.
- Trim extra spaces and make comparisons case-insensitive where appropriate.
- Keep town and taluka names consistent across registration and search.
- Let providers list multiple service areas if they travel beyond their registered town.
- Display the provider's service area clearly so customers know where they operate.

9. Security and correctness

- Use prepared SQL statements for search filters and submitted form values.
- Escape displayed values with htmlspecialchars().
- Validate all fields on the server, not only in the browser.
- Hash passwords using password_hash() and verify with password_verify().
- Use CSRF tokens for state-changing forms.
- Enforce customer, provider, and administrator roles on every protected action.
- Verify ownership before reading or changing private bookings or provider listings.
- Recheck service activity and provider approval when a booking is submitted.

- Use foreign keys and a unique constraint on email.
- Keep SQL table and column names consistent across every PHP page.

10. How to test the search

- Register a provider in Kolhapur district, Karveer taluka, Kolhapur town.
- Approve the provider through the administrator account.
- Add an active Electrical service for that provider.
- Open services.php and select Electrical, Kolhapur, Karveer, and Kolhapur.
- Confirm that the listing appears and displays the provider's location.
- Create another provider in a different district and verify the district filter excludes them.
- Clear the town filter and confirm results broaden while remaining filters still apply.
- Deactivate a service and confirm it no longer appears.
- Set a provider to pending or blocked and confirm their listings disappear.
- Test empty results and make sure one user cannot access another user's bookings.

11. Possible improvements

- Dependent district/taluka/town dropdowns backed by location tables.
- Multiple service areas for each provider.
- Provider profile pages with experience and contact details.
- Verified reviews after completed bookings.
- Booking notifications and status history.
- Map or GPS search after coordinate data and distance logic are implemented.

Final explanation

FixServices finds local services by storing each provider's location and service listings, receiving a customer's selected location filters, and querying active services from approved providers in those areas. The customer chooses a listing and requests a booking; the provider responds in their dashboard; and the customer tracks progress. Accurate, consistently recorded district, taluka, town, and service-area data is what makes the local search reliable.